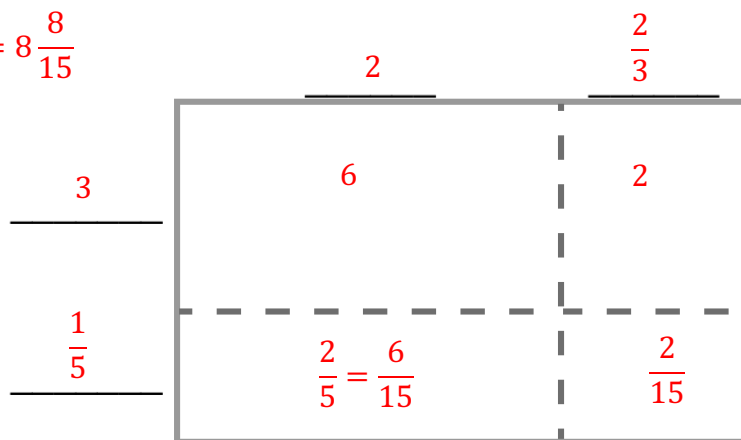


Grade 6 Accelerated
Unit 1
Lesson 6 Homework

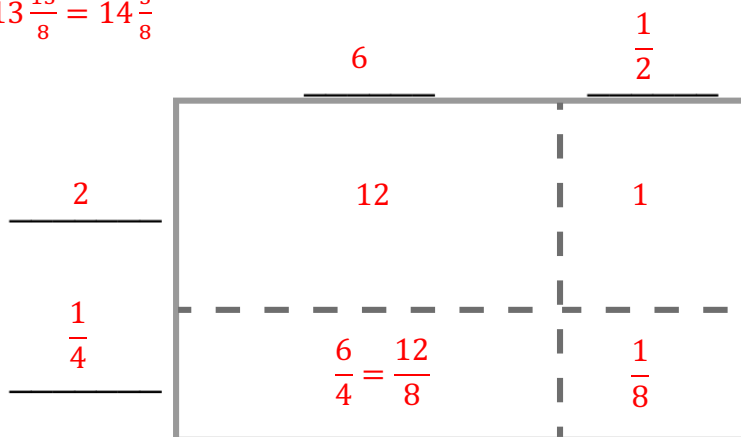
Name Answer Key
Date _____
Period _____

Find each product using the area model.

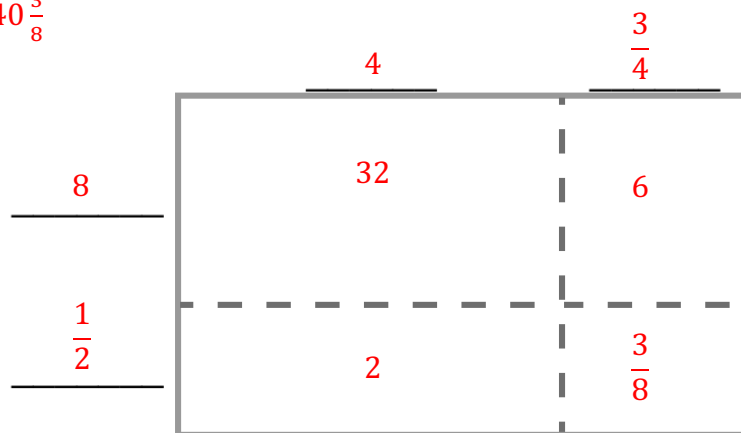
1. $2\frac{2}{3} \times 3\frac{1}{5} = 8\frac{8}{15}$



2. $6\frac{1}{2} \times 2\frac{1}{4} = 13\frac{13}{8} = 14\frac{5}{8}$



3. $4\frac{3}{4} \times 8\frac{1}{2} = 40\frac{3}{8}$



Find each product. Rewrite each product in simplest form. Show your work.

$$4. \quad \frac{7}{20} \times \frac{2}{14} = \frac{1}{20}$$

$$= \frac{14}{280} = \frac{1 \times \cancel{14}}{20 \times \cancel{14}} = \frac{1}{20}$$

$$5. \quad \frac{5}{12} \times 4\frac{1}{2} = 1\frac{7}{8}$$

$$= \frac{5}{12} \times \frac{9}{2} = \frac{45}{24} = \frac{15 \times \cancel{3}}{8 \times \cancel{3}} = \frac{15}{8} = 1\frac{7}{8}$$

$$6. \quad 3\frac{5}{8} \times 2\frac{3}{5} = 9\frac{17}{40}$$

$$= \frac{29}{8} \times \frac{13}{5} = \frac{377}{40} = 9\frac{17}{40}$$

$$7. \quad \frac{21}{100} \times 2\frac{2}{3} = \frac{14}{25}$$

$$= \frac{21}{100} \times \frac{8}{3} = \frac{168}{300} = \frac{14 \times \cancel{12}}{25 \times \cancel{12}} = \frac{14}{25}$$

$$8. \quad 8\frac{7}{15} \times \frac{3}{4} = 6\frac{7}{20}$$

$$= \frac{127}{15} \times \frac{3}{4} = \frac{381}{60} = \frac{127 \times \cancel{3}}{20 \times \cancel{3}} = \frac{127}{20} = 6\frac{7}{20}$$

$$9. \quad 5\frac{18}{25} \times 3\frac{1}{2} = 20\frac{1}{50}$$

$$= \frac{143}{25} \times \frac{7}{2} = \frac{1001}{50} = 20\frac{1}{50}$$

10. Arlo and Breanna were working on the multiplication expression $4\frac{5}{9} \times 3\frac{2}{5}$. Arlo says that the product is $12\frac{10}{45}$ or $12\frac{2}{9}$ because $4 \times 3 = 12$ and $\frac{5}{9} \times \frac{2}{5} = \frac{10}{45}$. Breanna says that the product is $\frac{697}{45}$ or $15\frac{22}{45}$ because $\frac{41}{9} \times \frac{17}{5} = \frac{697}{45}$. Explain who is correct and the mistake that the other student made.

Breanna is correct because you can't separate the whole numbers from the fractions when multiplying mixed numbers. You have to put them in improper fractions, like Breanna, before multiplying them.